

I2C	Sub SYS	Function	Part Number	I2C Spec.		I2C Slave Address / Write / Read (7-bit mode)	
I2C-0	AP	Flash LED driver	AW36413			0x6B	
I2C-1 (I3C)	AP Sensor Hub	Magnetic Sensor	AK09918C	400 Kbps		0x0C	Write:0x18 / Read:0x19
		Ambient Light Sensor	STK33562	400 Kbps		0x46	Write:0x8C / Read:0x8D
		Proximity Sensor					
		SAR Sensor	SX9328	400 Kbps		0x28	
I2C-2 (I3C)	AP	Rear Camera					
I2C-3	AP	LCM BIAS	SM5109C			0x3E	Write:0x7C / Read:0x7D
		Audio Smart PA	AW87339				
I2C-4 (I3C)	AP	Front Camera					
I2C-5	AP	Charger IC	BQ25601	400Kbps		0x6B	Write:0xD6 / Read:0xD7
I2C-6	AP	BUCK LP4X					

Note : I2C Spec. : Standard mode (100 kbps) and Fast mode (400 kbps), Fast mode Plus (1 Mbps) and High-speed mode (3.4 Mbps)

华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 02_I2C_ID_Overview

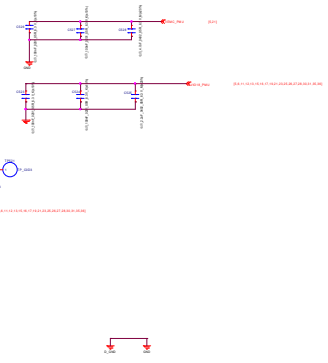
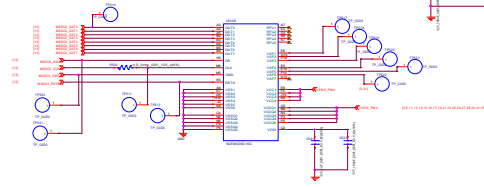
Size C	Project PA5905_MB_V1	Rev v1
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Date: Monday, January 01, 2024	Sheet 2 of 11
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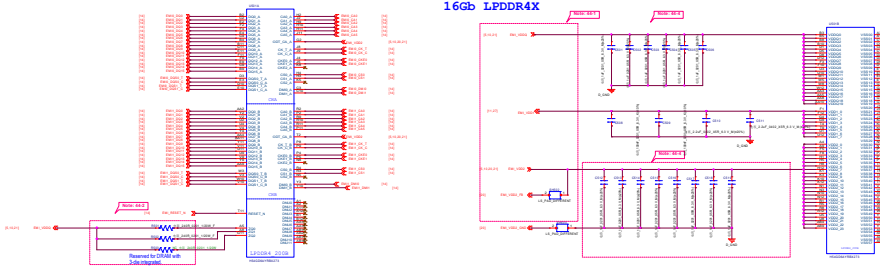
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 03_Change_Notice		
Size C	Project PA5905_MB_V1	Rev V1
Date: Monday, January 01, 2024		Sheet 3 of 11

华勤通讯 Huaqin Telecom Technology Com., Ltd		
Title 04_CHANGELIST		
Size D	Project PA5905_MB_V1	Rev V1
Date: Monday, January 01, 2024		Sheet 4 of 11

emmc 5.1

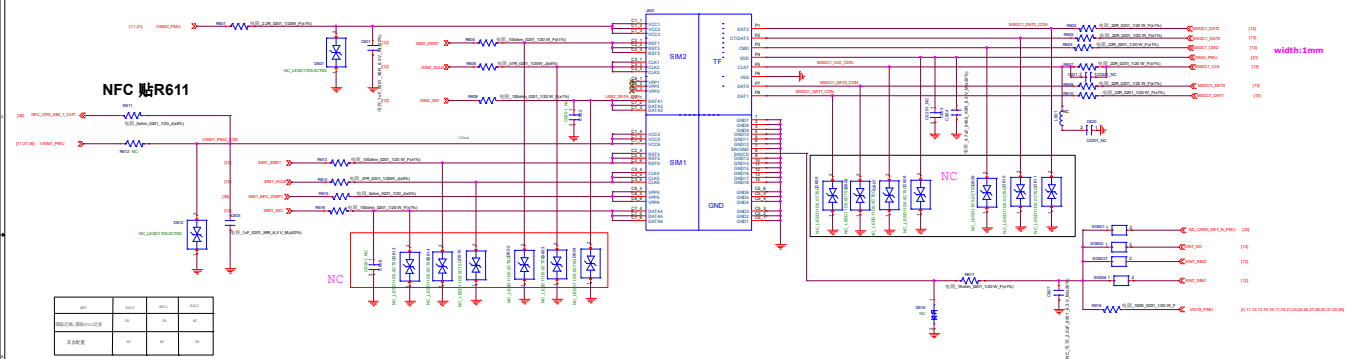


discrete LP4X 16Gb LPDDR4X



华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 05_Memory_eMMC_LPDDR4		
Size E	Project AL7015A_MB_V1	Rev v1
Date: Monday, January 01, 2024 Sheet 5 of 36		

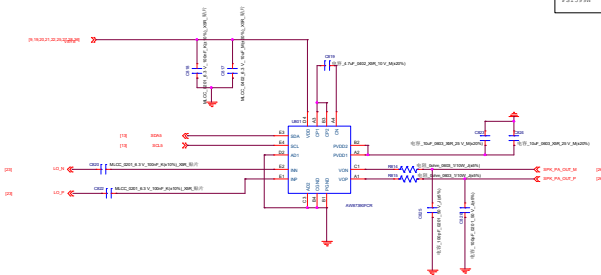
3选2卡座



0011	0010	0011	0012
国际之间:国际=0011之间	000	000	002
其余配置	001	002	000

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 06_Memory_SD_SIM_Card			
Size D	Project AL7015A_MB_V1		Rev v1
Date: Monday, January 01, 2024		Sheet 6 of 36	

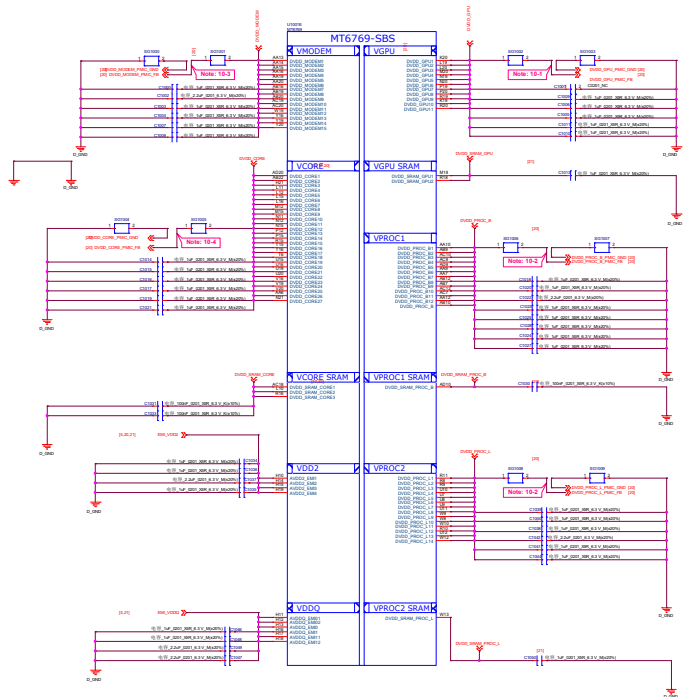
SPEAKER



01	02
03	04
05	06

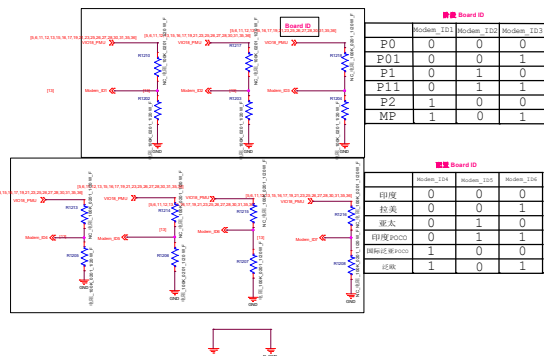
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 08_AUDIO_SPEAKER_AMP		
Size D	Project AL7015A_MB_V1	Rev v1
Date: Monday, January 01, 2024		Sheet 8 of 36

Schematic design notice of "10_BB_POWER_PDN" page:



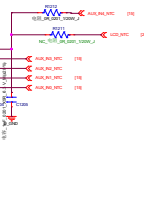
- Note 10-1: Differential pair of DVDD_GPU remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMC (Differential)
- Note 10-2: Differential pair of DVDD_PROC remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMC (Differential)
- Note 10-3: Differential pair of DVDD_MODEM remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMC (Differential)
- Note 10-4: Differential pair of DVDD_CORE remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMC (Differential)

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 10_BB_POWER_PDN		
Size D	Project AL7015A_MB_V1	Rev V1
Date: Monday, January 01, 2024	Sheet 10 of 36	



設定 Board ID			
	Modem_ID1	Modem_ID2	Modem_ID3
P0	0	0	0
P01	0	0	1
P1	0	1	0
P11	0	1	1
P2	1	0	0
MP	1	0	1

	Modern_104	Modern_1215	Modern_1242
印度	0	0	0
拉美	0	0	1
亚太	0	1	0
印度 POCO	0	1	1
国际轻奢 POCO	1	0	0
泛欧	1	0	1



Note 12-1: PWRAP, SPI0_CSN" and "AUD_DAT_MOSI0" are bootstrap pins to select which interface will be the JTAG pin out.

PWPRAP SPI0 CSN	AUD DAT MOSIO	JTAG Function	
default=PU	default=PD	AP JTAG	MD JTAG
HI	LO	N/A	N/A
HI	HI	SPI0 + EINT8	SPI2 + SPI3
LO	LO	SPI0 + EINT8	N/A
LO	HI	N/A	N/A

Note 12-2: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

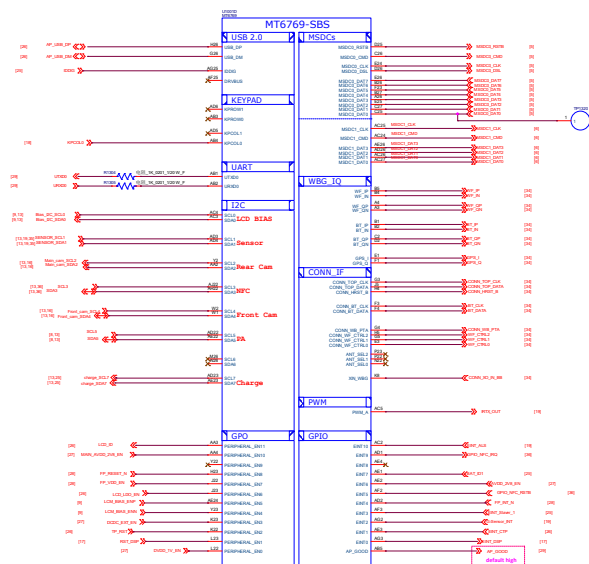
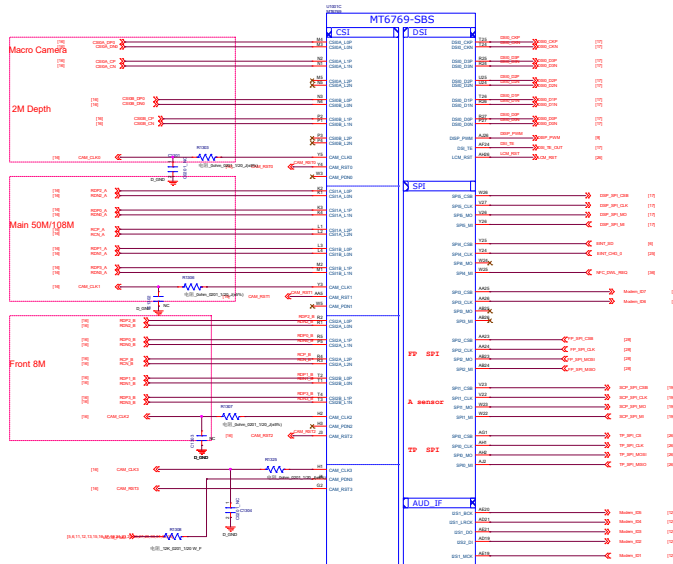
Note 12-3: The de-coupling cap. for REFP (AJ18 ball) have to be placed as close to BB as possible

Note 12-4: HW pin for DDR type feature in bootstrap (refer to BB HW design Notice)

AUD_SYNC_MISO	AUD_CLK_MISO	CAM_P0N3	PMC_6356 voltage	D0R Type
default=PD	default=PD	default=PD	VDRAM1 / VDRAM2	D0R
No ext. pull	No ext. pull	No ext. pull	1.25v / 0.6v	LP4X eMCP
No ext. pull	12K pull to VIO18	No ext. pull	OFF / 1.8v	Reserved
12K pull to VIO18	No ext. pull	No ext. pull	1.225v / OFF	LP3 eMCP
12K pull to VIO18	12K pull to VIO18	No ext. pull	1.25v / 1.8v	Reserved

Note 12-5: Charger must have D+/D- pin for charger type USB detection.
Charger must have at least 500mA USB current for All charger type

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 09_LCD_CTP_BL			
Size D	Project PA5905_MB_V1		Rev v1
Date: Monday, January 01, 2024		Sheet 12 of 36	

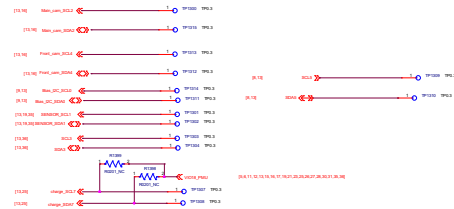


Design Notice: Boot mode

MT6768/67 DDR trapping table

	MT6768 / MT6767	MT6768 / MT6767	CAPI_FTM3 (Not set)	PMIC MT6358	
	AUX_SYNC_MISO (Bit 628)	AUX_SYNC_MISO (Bit 628)	AUX_SYNC_MISO (Bit 628)	VDRAM1	VDRAM2
Reserved	No ext. pull	No ext. pull	No ext. pull	3.125V	0.6V
Reserved	No ext. pull	12K to VIO18	No ext. pull	0V	1.8V
Reserved	12K to VIO18	No ext. pull	No ext. pull	3.125V	0V
Reserved	12K to VIO18	12K to VIO18	No ext. pull	3.125V	1.8V
Reserved	No ext. pull	No ext. pull	12K to VIO18	3.125V	0.6V
Reserved	No ext. pull	12K to VIO18	12K to VIO18	0V	1.8V
Reserved	12K to VIO18	No ext. pull	12K to VIO18	3.125V	0V
Reserved	12K to VIO18	12K to VIO18	12K to VIO18	3.125V	1.8V

I2C TEST

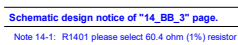


华勤通讯 Huaqin Telecom Technology Com., Ltd

Title 13_BB2

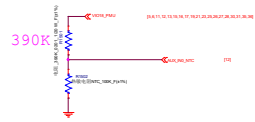
Size D Project AL7015A_MB_V1 Rev V1

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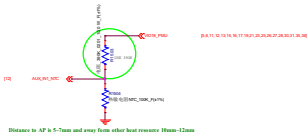
华勤通讯 Huaqin Telecom Technology Co.,Ltd		
Title 09_LCD_CTP_BL		
Size D	Project PA5905_MB_V1	Rev v1
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CPU_THERMAL



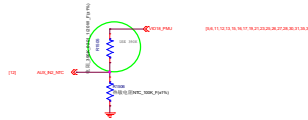
change to NCP03WF104F05RL Murata

4G_PA_THERMAL



Distance to AP is 5-7mm and away from other heat sources (Hsiao-Tsuan)

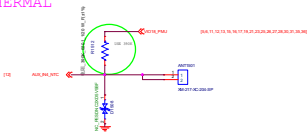
charger



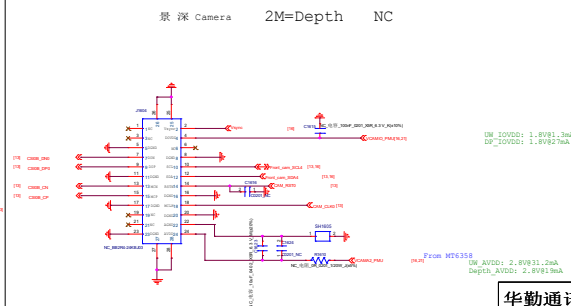
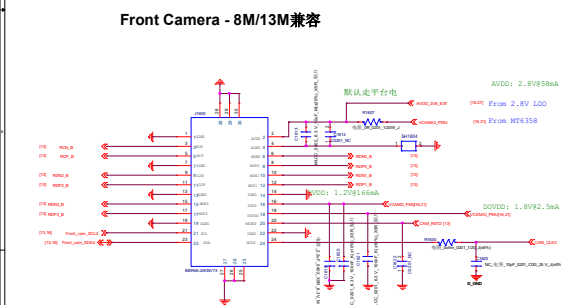
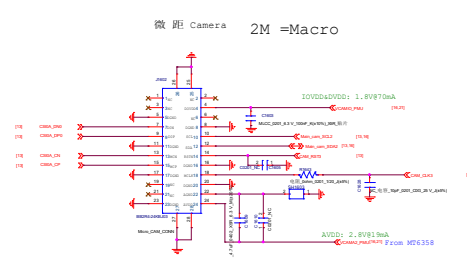
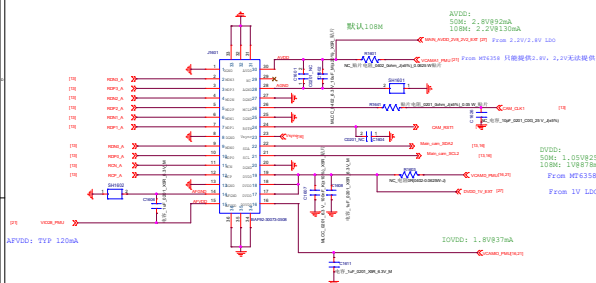
BOARD_THERMAL



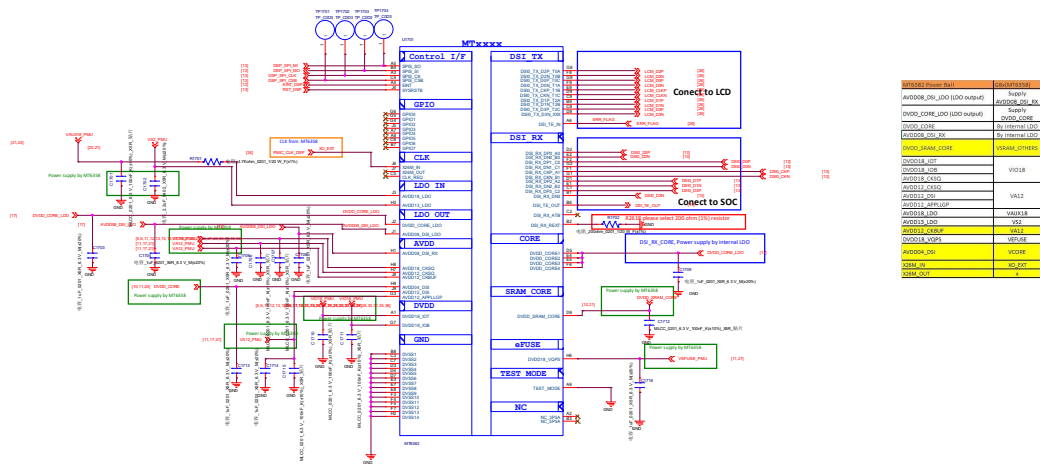
FlashLED_THERMAL



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 15_BB_AUXADC_Thermal			
Size D	Project AL7015A_MB_V1		Rev v1
Date: Monday, January 01, 2024		Sheet 15 of 36	

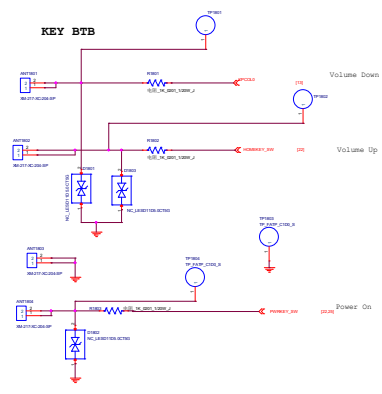


华勤通讯 Huaqin Telecom Technology Co.,Ltd		
Title 16_CAMERA_INTERFACE		
Size D	Project AL7015A_MB_V1	Rev V1
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Pin	Signal	IO Type
AVDD08_08_100 (IO0 output)	IO0[0]	IO
AVDD08_08_100 (IO0 output)	IO0[1]	IO
AVDD08_08_100 (IO0 output)	IO0[2]	IO
AVDD08_08_100 (IO0 output)	IO0[3]	IO
AVDD08_08_100 (IO0 output)	IO0[4]	IO
AVDD08_08_100 (IO0 output)	IO0[5]	IO
AVDD08_08_100 (IO0 output)	IO0[6]	IO
AVDD08_08_100 (IO0 output)	IO0[7]	IO
AVDD08_08_100 (IO0 output)	IO0[8]	IO
AVDD08_08_100 (IO0 output)	IO0[9]	IO
AVDD08_08_100 (IO0 output)	IO0[10]	IO
AVDD08_08_100 (IO0 output)	IO0[11]	IO
AVDD08_08_100 (IO0 output)	IO0[12]	IO
AVDD08_08_100 (IO0 output)	IO0[13]	IO
AVDD08_08_100 (IO0 output)	IO0[14]	IO
AVDD08_08_100 (IO0 output)	IO0[15]	IO
AVDD08_08_100 (IO0 output)	IO0[16]	IO
AVDD08_08_100 (IO0 output)	IO0[17]	IO
AVDD08_08_100 (IO0 output)	IO0[18]	IO
AVDD08_08_100 (IO0 output)	IO0[19]	IO
AVDD08_08_100 (IO0 output)	IO0[20]	IO
AVDD08_08_100 (IO0 output)	IO0[21]	IO
AVDD08_08_100 (IO0 output)	IO0[22]	IO
AVDD08_08_100 (IO0 output)	IO0[23]	IO
AVDD08_08_100 (IO0 output)	IO0[24]	IO
AVDD08_08_100 (IO0 output)	IO0[25]	IO
AVDD08_08_100 (IO0 output)	IO0[26]	IO
AVDD08_08_100 (IO0 output)	IO0[27]	IO
AVDD08_08_100 (IO0 output)	IO0[28]	IO
AVDD08_08_100 (IO0 output)	IO0[29]	IO
AVDD08_08_100 (IO0 output)	IO0[30]	IO
AVDD08_08_100 (IO0 output)	IO0[31]	IO
AVDD08_08_100 (IO0 output)	IO0[32]	IO
AVDD08_08_100 (IO0 output)	IO0[33]	IO
AVDD08_08_100 (IO0 output)	IO0[34]	IO
AVDD08_08_100 (IO0 output)	IO0[35]	IO
AVDD08_08_100 (IO0 output)	IO0[36]	IO
AVDD08_08_100 (IO0 output)	IO0[37]	IO
AVDD08_08_100 (IO0 output)	IO0[38]	IO
AVDD08_08_100 (IO0 output)	IO0[39]	IO
AVDD08_08_100 (IO0 output)	IO0[40]	IO
AVDD08_08_100 (IO0 output)	IO0[41]	IO
AVDD08_08_100 (IO0 output)	IO0[42]	IO
AVDD08_08_100 (IO0 output)	IO0[43]	IO
AVDD08_08_100 (IO0 output)	IO0[44]	IO
AVDD08_08_100 (IO0 output)	IO0[45]	IO
AVDD08_08_100 (IO0 output)	IO0[46]	IO
AVDD08_08_100 (IO0 output)	IO0[47]	IO
AVDD08_08_100 (IO0 output)	IO0[48]	IO
AVDD08_08_100 (IO0 output)	IO0[49]	IO
AVDD08_08_100 (IO0 output)	IO0[50]	IO
AVDD08_08_100 (IO0 output)	IO0[51]	IO
AVDD08_08_100 (IO0 output)	IO0[52]	IO
AVDD08_08_100 (IO0 output)	IO0[53]	IO
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AVDD08_08_100 (IO0 output)	IO0[61]	IO
AVDD08_08_100 (IO0 output)	IO0[62]	IO
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AVDD08_08_100 (IO0 output)	IO0[94]	IO
AVDD08_08_100 (IO0 output)	IO0[95]	IO
AVDD08_08_100 (IO0 output)	IO0[96]	IO
AVDD08_08_100 (IO0 output)	IO0[97]	IO
AVDD08_08_100 (IO0 output)	IO0[98]	IO
AVDD08_08_100 (IO0 output)	IO0[99]	IO

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title	17_bridge IC	
Size D	Project AL7015A_MB_V1	Rev V1
Date: Monday, January 01, 2024	Sheet 17 of 36	



Schematic design notice of "65_PERI_Dual_SIM_ICUSB_KEYPAD" page.

Note 75-1: DO NOT put pull-up resistor on PWRKEY

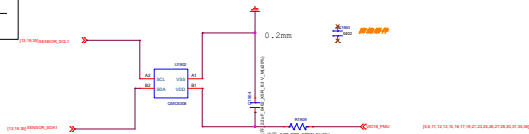
Note 75-2: Volume Up : HOME Key / GND
Volume Down : KPROW0/KPCOL0

Note 75-3: For better ESD performance, please select suitable components for system protection.

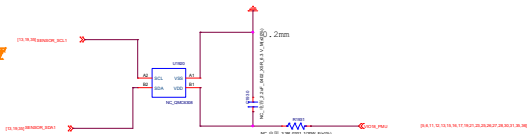
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 18_KEYPAD_INTERFACE		
Size D	Project AL7015A_MB_V1	Rev v1
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eCOMPASS

型号	封装
QMC6308	DSOC
HMCS60 DS05	DS05

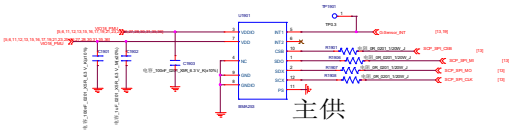


预留位置



G-Sensor

型号	封装
SGS8253	DS18
SGS8122	DS0F

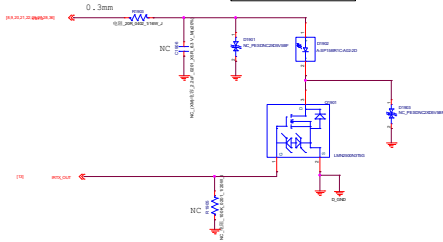


主供

兼容二供

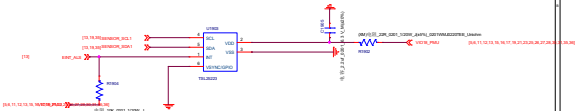
IR LED

厂家	型号
Everlight 威盛	2826-61C-4510-B-02
光磊	A-28158181C-A02-02



ALS

	I2C ADDRESS
TEL25133	0x39



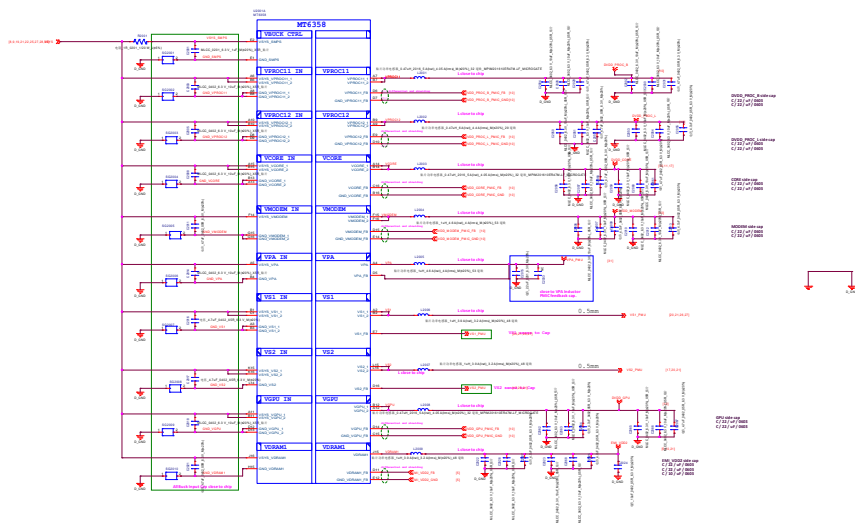
华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 19_SENSORS_ALS_PS_A

Size D Project AL7015A_MB_V1

Rev v1

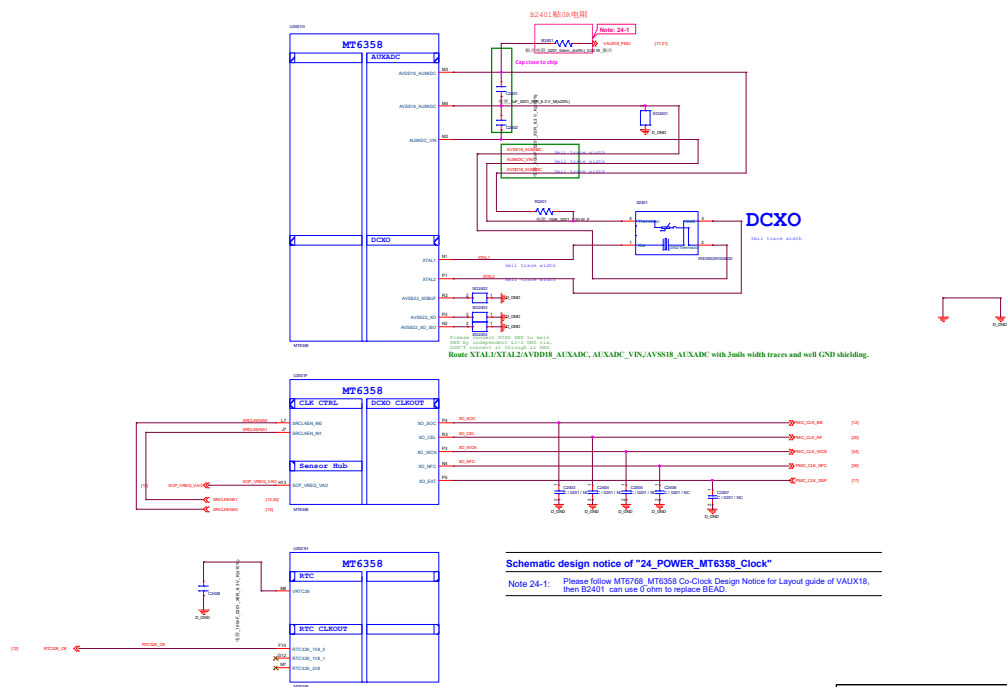
Date: Monday, January 01, 2024 Sheet 19 of 36



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 20_POWER_MT6358-Buck			
Size E	Project	AL7015A_MB_V1	Rev v1
Date: Monday, January 01, 2024		Sheet 20 of 40	

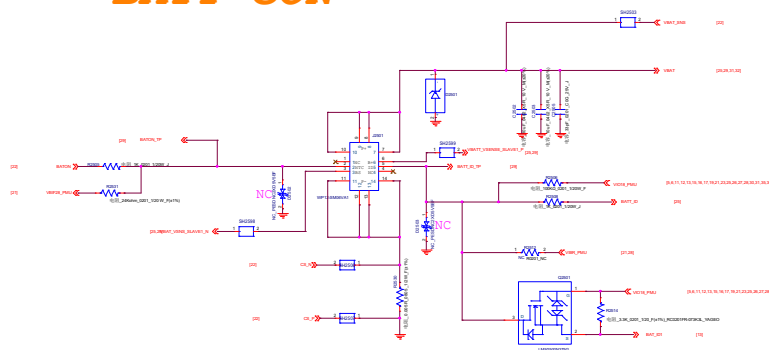


华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 22_POWER_MT6358-IF		
Size D	Project AL7015A_MB_V1	Rev v1
Date: Monday, January 01, 2024	Sheet 22 of 36	

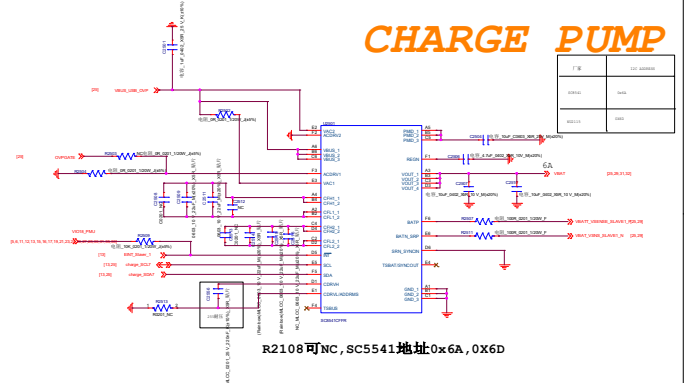


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		24_POWER_MT6358_Clock	
Size	Project	AL7015A_MB_V1	Rev
D			V1
Date:	Monday, January 01, 2024	Sheet	24 of 36

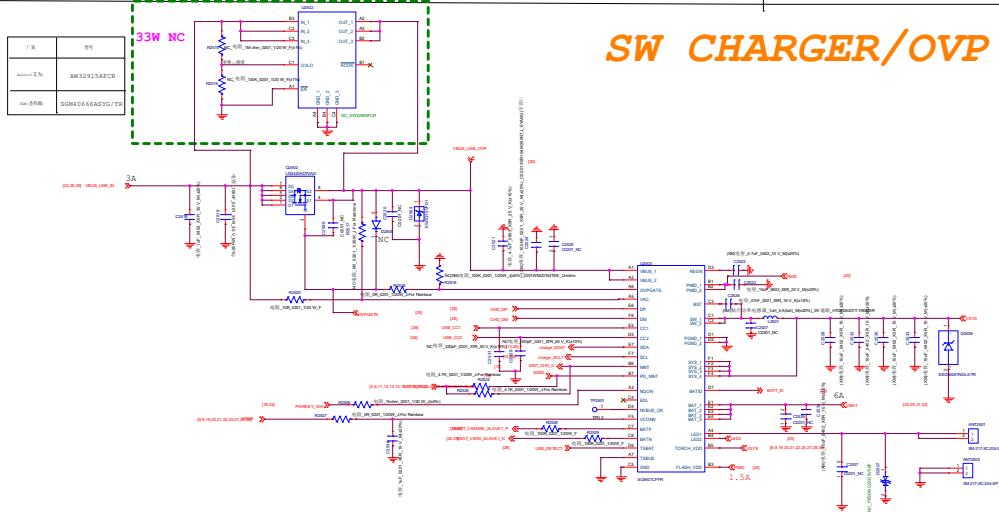
BATT CON



CHARGE PUMP



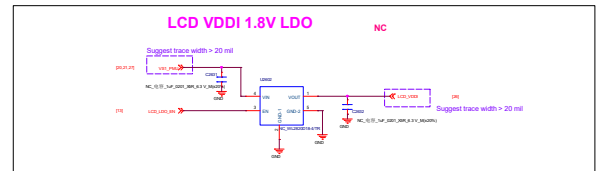
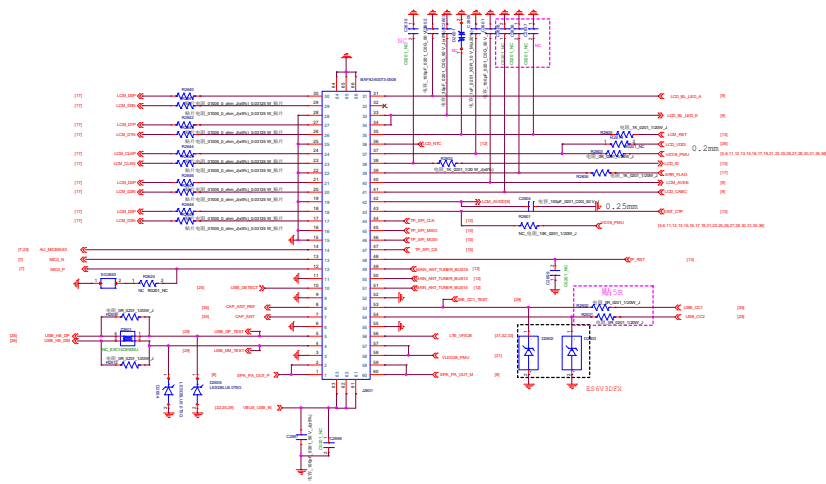
SW CHARGER/OVP



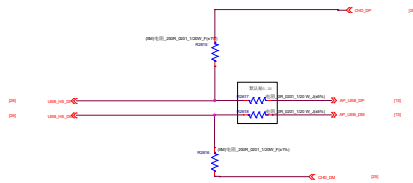
闪光灯TVS NC

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 26_POWER_SubPMIC-Charger+CONN			
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LINK

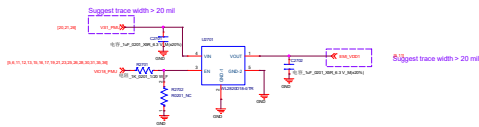


DP / DM



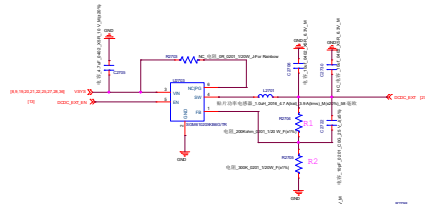
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 26_LCD 60PIN			
Size D	Project AL7015A_MB_V1		Rev v1
Date: Thursday, February 01, 2024		Sheet 26 of 36	

LPDDR4X VDD1 1.8V LDO Note: 28-5



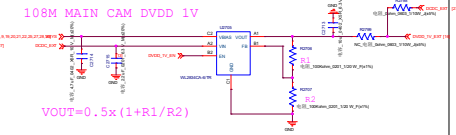
1.8V	1.8V
1.8V	1.8V
1.8V	1.8V
1.8V	1.8V

108M CAM DCDC EXT VOUT=0.6x(1+R1/R2)

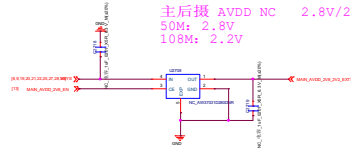


108M MAIN CAM DVDD 1V

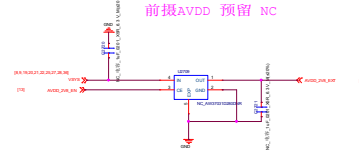
$$VOUT=0.5 \times (1+R1/R2)$$



主后摄 AVDD NC 2.8V/2.2V封装一致
50M: 2.8V
108M: 2.2V



前摄AVDD 预留 NC



Schematic design notice of "28_POWER_ThirdParty-Power"

Note 28-1: VA12 Layout placement please close to AP

Note 28-2: VS2 Buck Layout placement please close to PMIC_M76367

Note 28-3: VCN33 LDO Layout placement please close to M76631

华勤通讯 Huaqin Telecom Technology Com.,Ltd

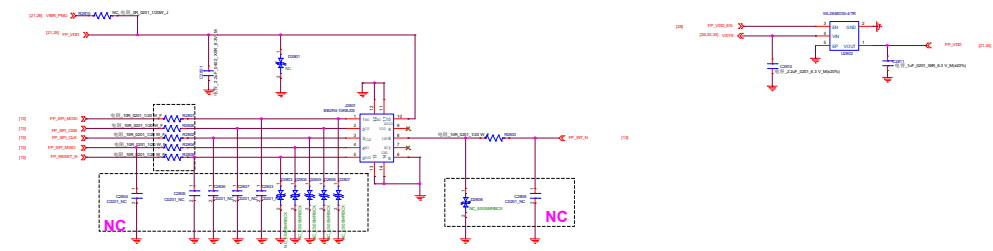
Title 27_POWER_ThirdParty_Powers

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Finger Print



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Title 28_Fingerprint		
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工厂测点



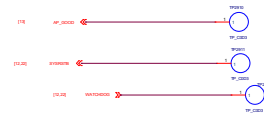
VBAT



串口



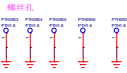
研发测点



屏蔽罩



光学定位孔

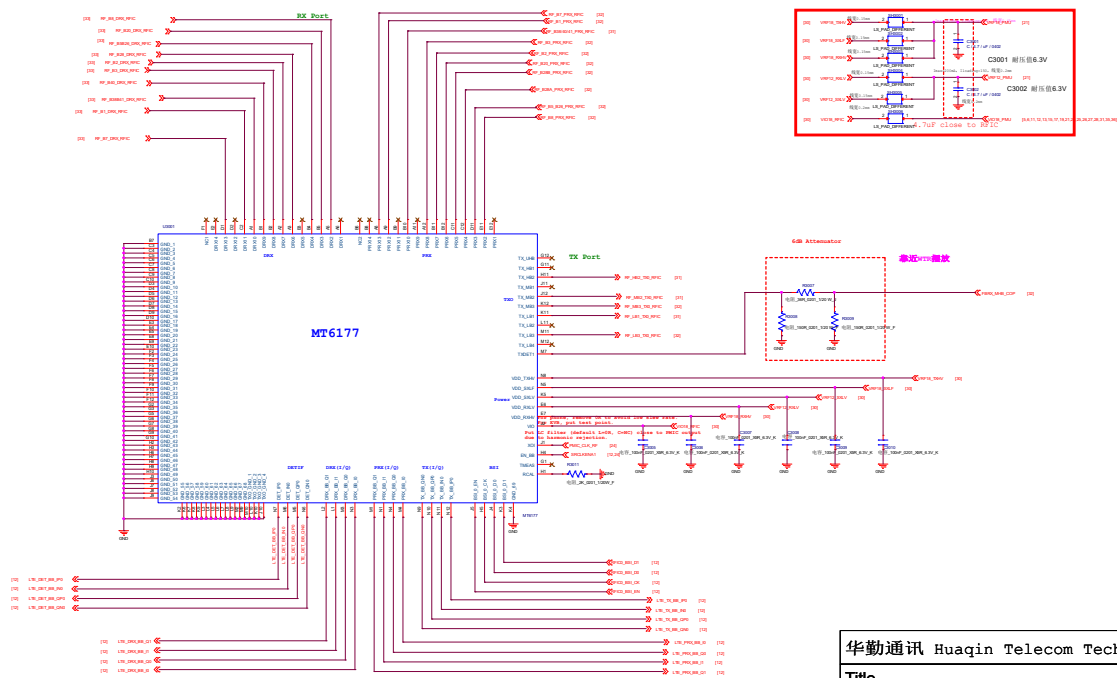


提升

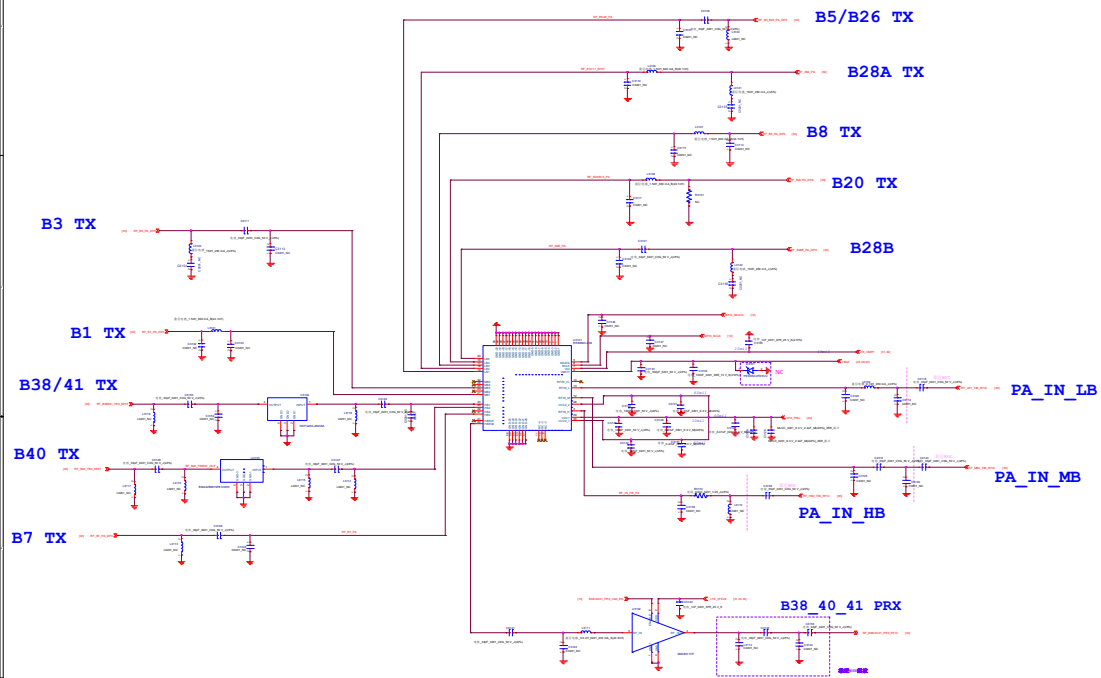


华勤通讯 Huaqin Telecom Technology Com., Ltd		
Title 29_DEBUG_IO		
Size D	Project AL7015A_MB_V1	Rev V1
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Power For MT6177

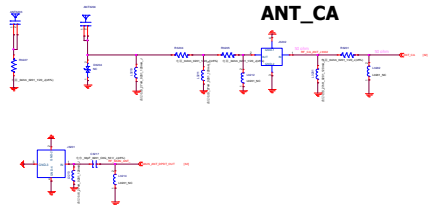


华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 31.RF_MT6186M_Transceiver		
Size D	Project PA9161_MB_V1	Rev v1
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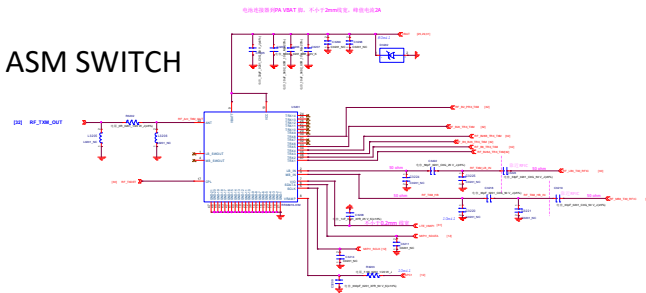


华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 32_RF_MT6177_RF_TX		
Size	Project	Rev
Custom	PA9161_MB_V1	v1
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ANT_CA



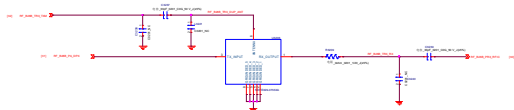
ASM SWITCH



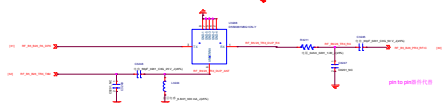
B2/PCS DPX



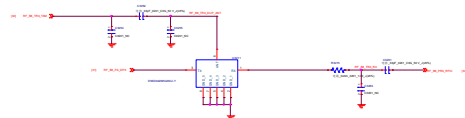
B28B TRX



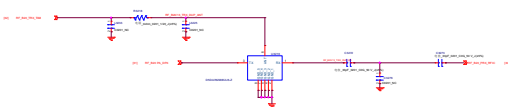
B5 DPX



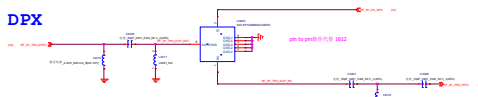
B8 DPX



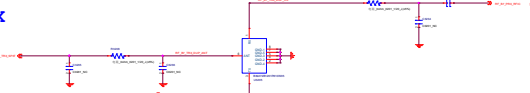
B20 DPX



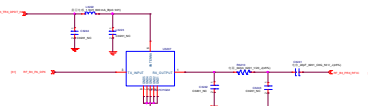
B1 DPX



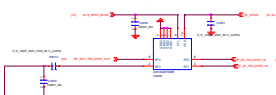
B7 PRX



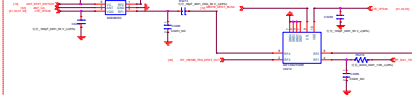
B3 DPX



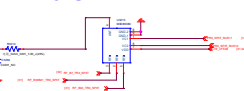
B1/3 DPDT



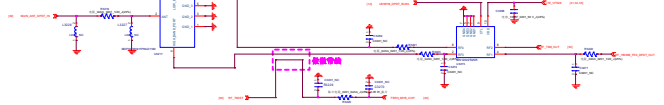
MB/HB DPDT



HB SP3T



LB/MHB DPDT

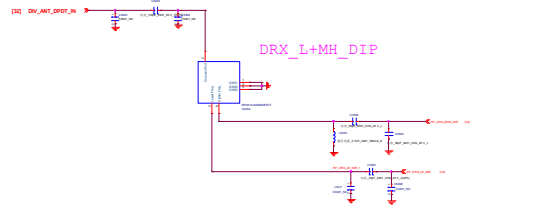
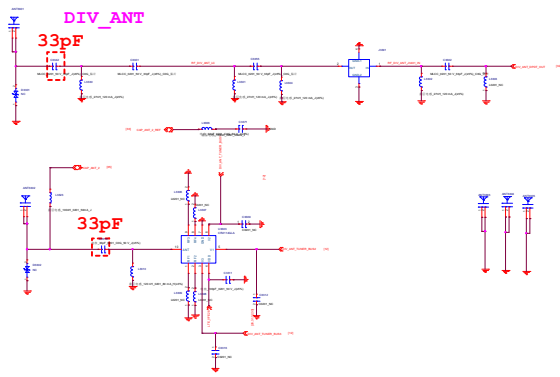


DPDT SWITCH

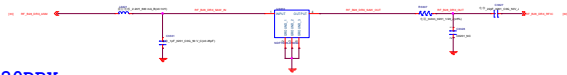


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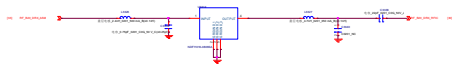
Rev v1



B28 DRX



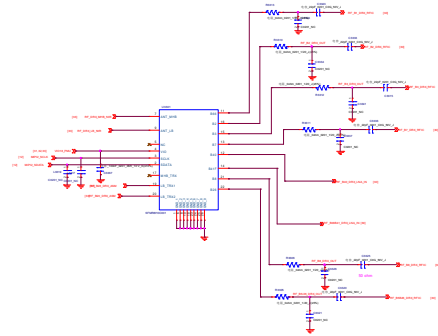
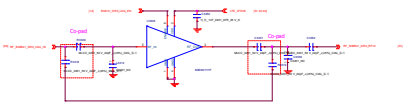
B20DRX



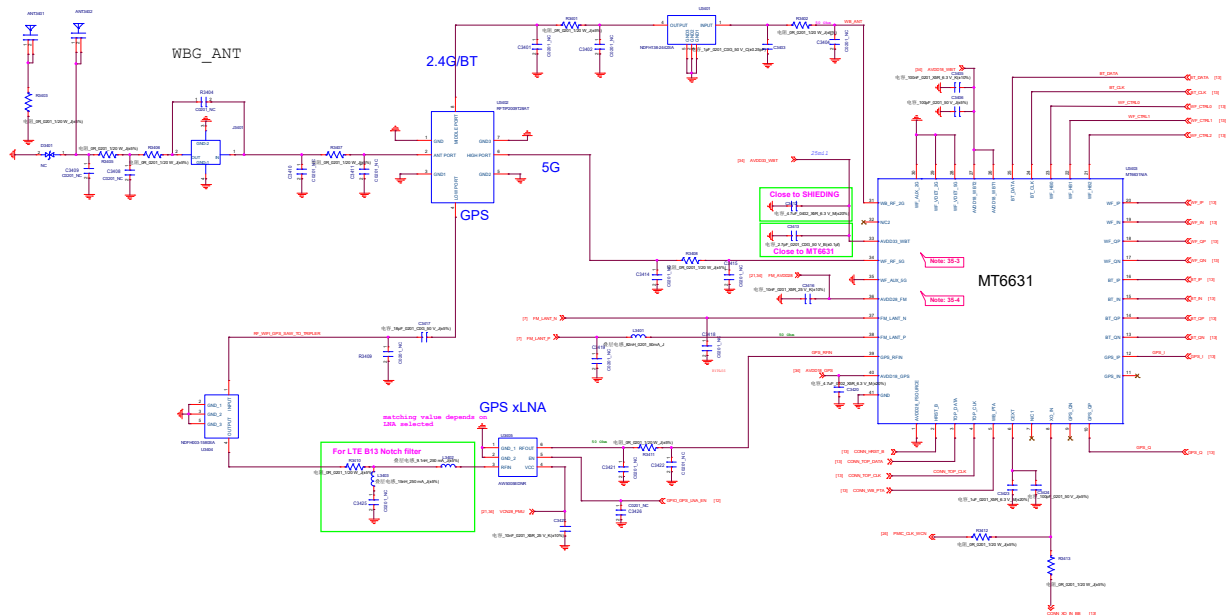
B40 DRX LNA



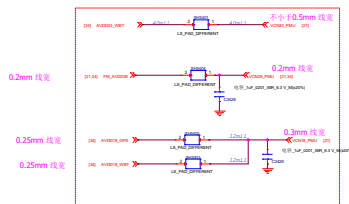
B38/41 DRX LNA



华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 34_RF_MT6177_RF_DRX		
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WCN Power Supply



Schematic design notice of "S1_CONNECTIVITY_CONSYS_MT6631"

Note 35-1: For R5015 size, please select 0402 size or larger one

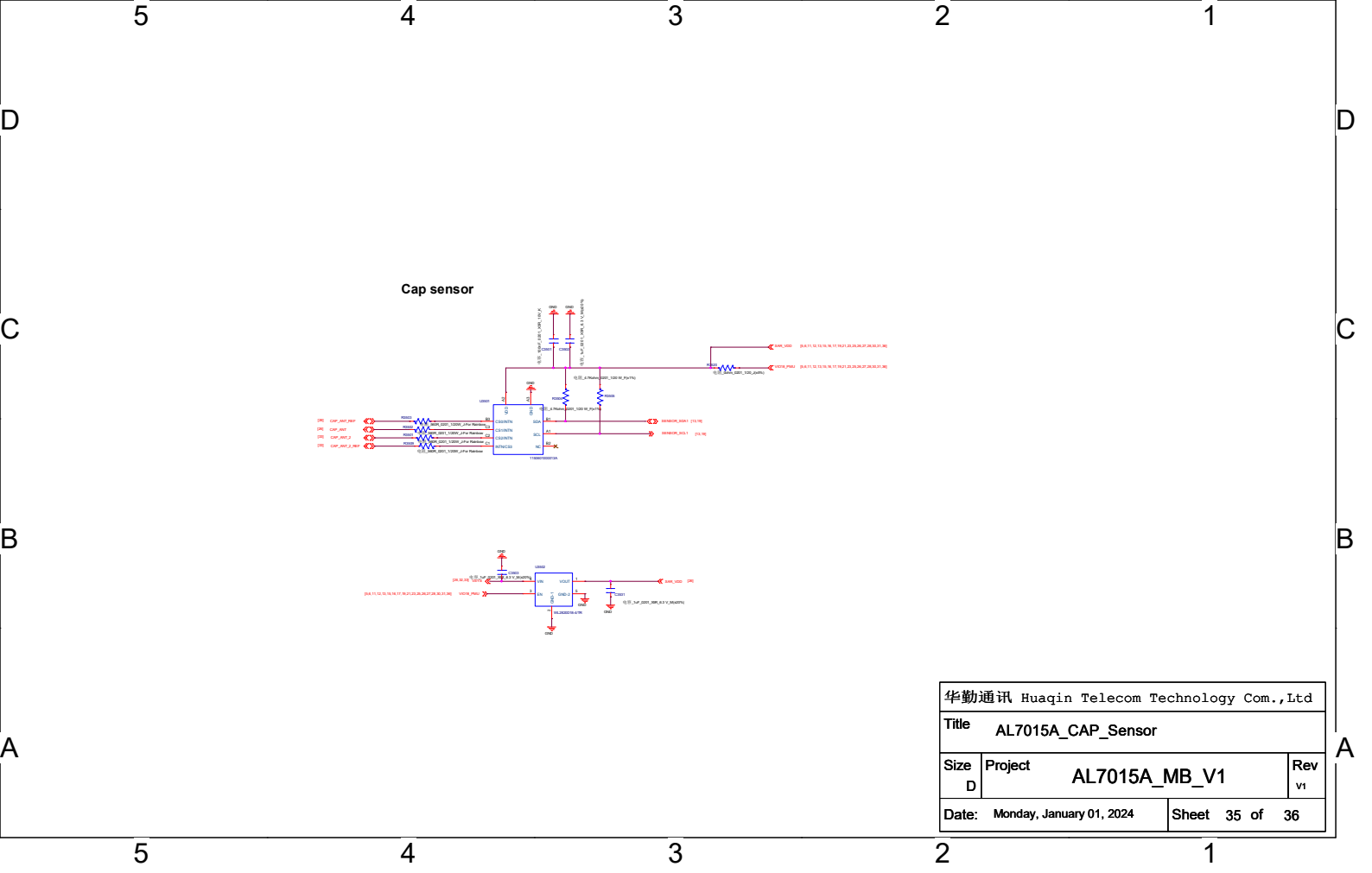
Note 35-2: Please refer to MT6631 Baseband design notice for VCN33 LDO selection guide

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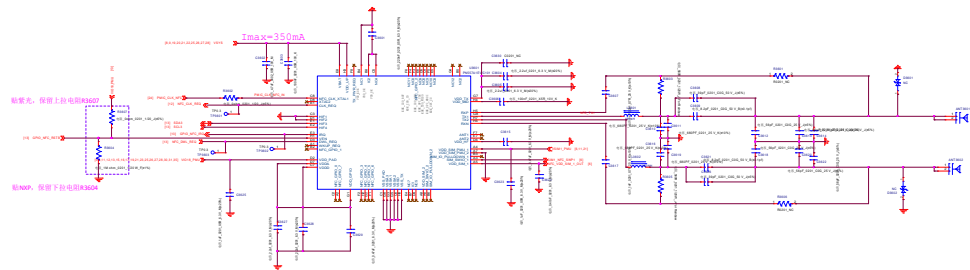
Title AL7015A_CONNECTIVITY_CONSYS_MT6631

Size D Project AL7015A_MB_V1 Rev V1

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Title AL7015A_NFC		
Size E	Project AL7015A_MB_V1	Rev v1
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